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Week 2: Camera Calibration

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Week 2 Quiz

Your grade: 100%

Next item

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

2.4 Intrinsic and Extrinsic Matrices Self-Check Quiz

Coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Submitted Attempted

Try again

Your grade

100%

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1. In camera calibration, what does the intrinsic matrix represent? 1 / 1 point

- The transformation from world coordinates to image coordinates
- The transformation from image coordinates to pixel coordinates
- The transformation from camera coordinates to image coordinates
- The transformation from pixel coordinates to image coordinates

Correct
In computer vision and camera calibration, the intrinsic matrix is a fundamental component of the camera model. The intrinsic matrix is essential for mapping 3D points in the camera coordinate system to 2D points on the camera sensor or the image plane. It also provides information about the camera's field of view, focal length, skew, and principal point, which are necessary for undistorting and transforming image points accurately.

2. True or False: The intrinsic matrix is the same for all images captured by a particular camera. 1 / 1 point

- True
- False

Correct
Since the intrinsic matrix represents the internal characteristics of the camera, we can conclude that an intrinsic matrix is specific to a particular camera and remains constant for all images captured by that camera.

3. Which of the following transformations are considered in the extrinsic matrix? (Choose all that apply.) 1 / 1 point

- [x] Translation along the x-axis

Correct
The extrinsic matrix accounts for how the camera is (A) translated and (B) rotated relative to the global coordinate system to capture the scene.

$$M_{ext} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- [x] Rotation around the y-axis

Correct
The extrinsic matrix accounts for how the camera is (A) translated and (B) rotated relative to the global coordinate system to capture the scene.

$$M_{ext} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- [ ] Scaling along the z-axis
- [ ] Shearing along the x-axis